# Parametric UMAP Trainer Implementation Plan

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The trainer will learn a reusable mapping from the six numerical features selected by AWRS–SMC to the two coordinates of the winning Common Lisp UMAP. It will not replace the search that constructed the reference atlas. Its purpose is to place a new study in that fixed atlas without fitting UMAP again.

Each selected feature is represented as a token. A token combines a learned feature-identity embedding with a learned projection of its standardized scalar value. A small Transformer encoder applies self-attention, residual connections, normalization, and a feed-forward block to the six tokens. Mean pooling followed by a regression head produces two coordinates. The entire encoder and regression head will be trained; this differs from the earlier pedagogical text Transformer, whose encoder was frozen while only its output heads were optimized.

The initial objective is coordinate mean squared error against the preserved Common Lisp coordinates. A later objective will add a neighbourhood term,

$$\mathcal{L} = \mathcal{L}_{\text{coordinate}} + \lambda \mathcal{L}_{\text{neighbourhood}},$$

where $\lambda$ controls the preference for preserving nearby observations. The coordinate-only model is retained as a baseline so that the effect of the neighbourhood term can be measured rather than assumed.

Implementation proceeds through reproducible milestones. First, a deliberately small model must overfit one corpus observation, be serialized as an S-expression, be loaded into a fresh process, and reproduce the same coordinates. Second, the model will be trained on the 70 training observations and evaluated on the 20 study-grouped validation observations. Third, prediction software will accept a new study's six features and return its location without rerunning UMAP. Finally, predicted coordinates will be assessed by coordinate error, nearest-neighbour retention, and cluster agreement with the winning atlas.

The pilot corpus is small, so the initial network will contain one attention block, one attention head, and a low-dimensional hidden representation. Increasing capacity will be justified only by validation results. The validation split does not independently validate the medical geometry: it measures how well the trainer approximates a fixed atlas for a held-out study group.